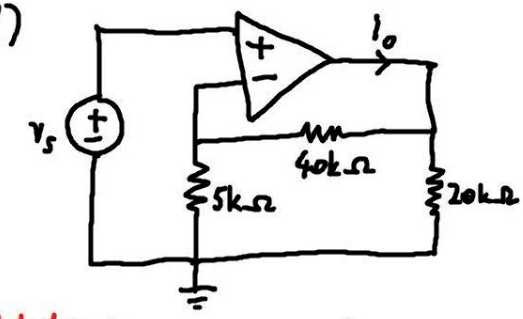
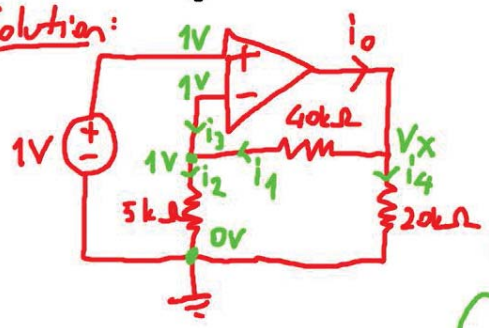
1)



Given $v_s = 1V$, find i_o.

Solution:



KCL: $i_1 + i_3 = i_2$
$i_3 = 0$ (Ideal Opamp) $\Big\} \Rightarrow i_1 = i_2$

Ohm's law: $1V = i_2 \cdot 5k\Omega \Rightarrow i_2 = 0{,}2mA \Rightarrow i_1 = 0{,}2mA$

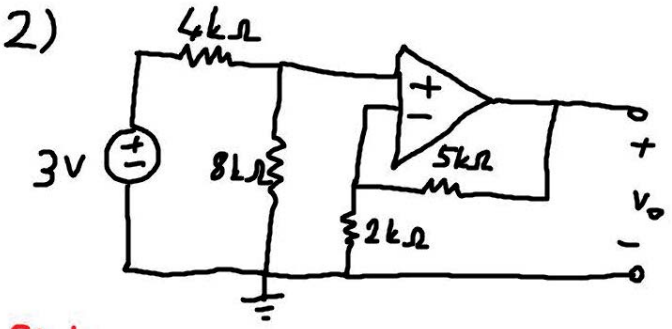
Ohm's law: $V_x - 1 = i_1 \cdot 40k\Omega = 0{,}2mA \cdot 40k\Omega = 8V \Rightarrow V_x = 9V$

KCL: $i_o = i_1 + i_4$

Ohm's law: $V_x = i_4 \cdot 20k\Omega \Rightarrow i_4 = \frac{V_x}{20k} = 0{,}45mA$

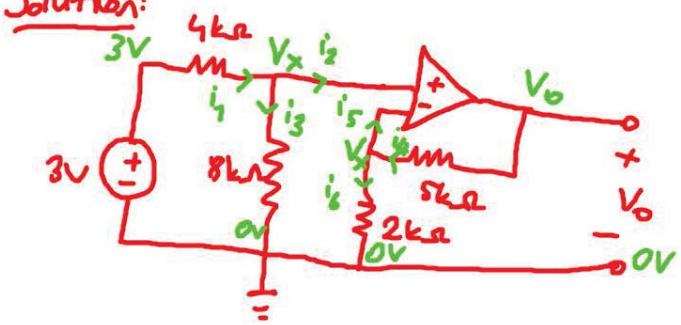
$\rightarrow i_o = 0{,}2 + 0{,}45 = 0{,}65mA$

2)



Find V_o.

Solution:



KCL: $i_1 = i_2 + i_3$ $\quad i_2 = 0$ (Ideal Opamp) $\Rightarrow i_1 = i_3$

Ohm's law: $3V - V_x = i_1 \cdot 4k\Omega$
Ohm's law: $V_x = i_3 \cdot 8k\Omega$ $\Big\} 6V - 2V_x = V_x$

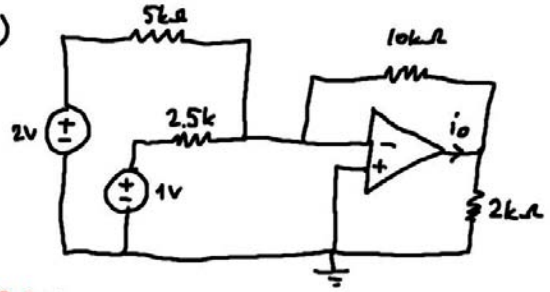
$V_x = 2V$

KCL: $i_4 = i_5 + i_6$ $\quad i_5 = 0$ (Ideal Opamp) $\Rightarrow i_4 = i_6$

Ohm's law: $V_o - V_x = i_4 \cdot 5k\Omega$
Ohm's law: $V_x = i_6 \cdot 2k\Omega$ $\Big\} 2V_o - 2V_x = 5V_x$

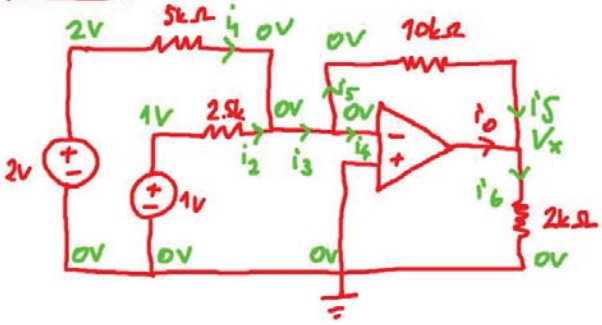
$V_o = \frac{7}{2} V_x = 7V$

3)



Find i_o.

Solution:



KCL: $i_1+i_2=i_3$ } $i_1+i_2=i_5$

KCL: $i_3=i_4+i_5 \Rightarrow i_3=i_5$

$i_4=0$ (Ideal Opamp)

Ohm's law: $2V=i_1\cdot 5k\Omega \Rightarrow i_1=0.4mA$

Ohm's law: $1V=i_2\cdot 2.5k\Omega \Rightarrow i_2=0.4mA$

$i_5=0.4+0.4$

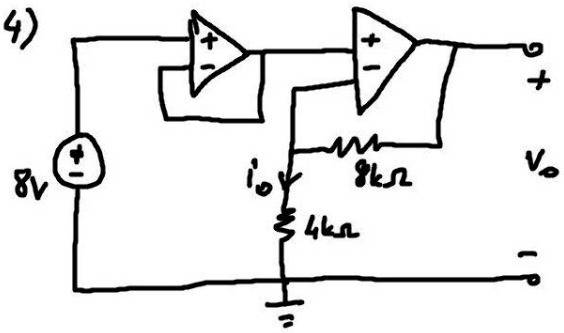
$=0.8mA$

Ohm's law: $-V_x=i_5\,10k\Omega \Rightarrow V_x=-8V$

Ohm's law: $V_x=i_6\cdot 2k\Omega \Rightarrow i_6=-4mA$

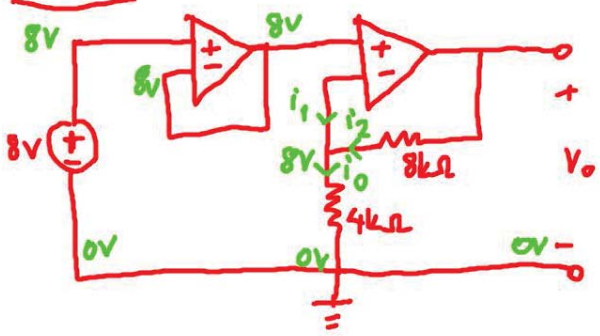
KCL: $i_6=i_o+i_5 \Rightarrow i_o=i_6-i_5=-4-0.8=-4.8mA$

4)



Find V_o and i_o.

Solution:



KCL: $i_1+i_2=i_o$ } $i_o=i_2$

$i_1=0$ (Ideal Opamp)

Ohm's law: $8V=i_o\cdot 4k\Omega \Rightarrow i_o=2mA$ } $V_o-8V=16V$

Ohm's law: $V_o-8V=i_2\cdot 8k\Omega$ } $V_o=24V$